

Endogenous Firm Scope via Organizational Complexity

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Scalable Capability, Bounded Scope

- **Intangibles** are an ever larger share of firms' capital, they are **non-rival**, and the largest firms rely on them most heavily. (Crouzet, Eberly, Eisfeldt & Papanikolaou 2024; Argente, Moreira, Oberfield & Venkateswaran 2025)
- **Firm scope** is **tight** and **related**: across a heavily right-skewed size distribution, mean scope varies by a factor of just 1.6, and firms expand into related markets. (Hoberg & Phillips 2025)
- Even Amazon (healthcare) and Google (hardware) **stall in distant markets**: why doesn't a handful of firms span the whole economy?

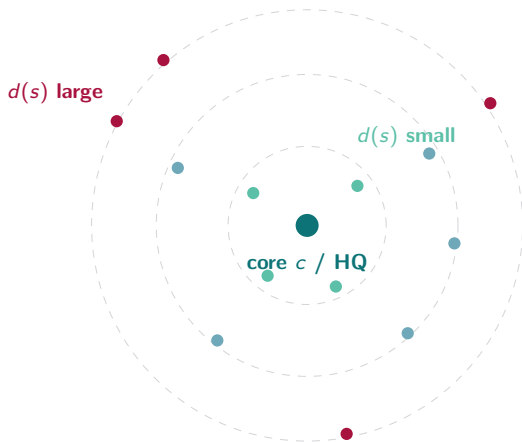
What **bounds firm scope?**

This Paper: Firm Scope under Organizational Complexity

- **Organizational complexity**: the cumulative heterogeneity of a firm's footprint, measured from its core.
 - the binding constraint on scope: how different the markets are, not how many
 - count-based alternatives: span of control (Lucas 1978), entry costs (Melitz 2003), knowledge hierarchies (Garicano 2000)
- **Scope**, the mass of markets the firm serves, is pinned down by two forces:
 - **non-separable** expansion: each added market raises the price of attention everywhere
 - **non-rival** blueprint: a larger footprint raises the return to improving it
 - firms expand along related lines and stop where complexity bites
- **Implications**:
 - **missing giants**: Pareto capability, exponentially bounded scope
 - **localization**: firms dominate their markets more, each firm **reaches less far**

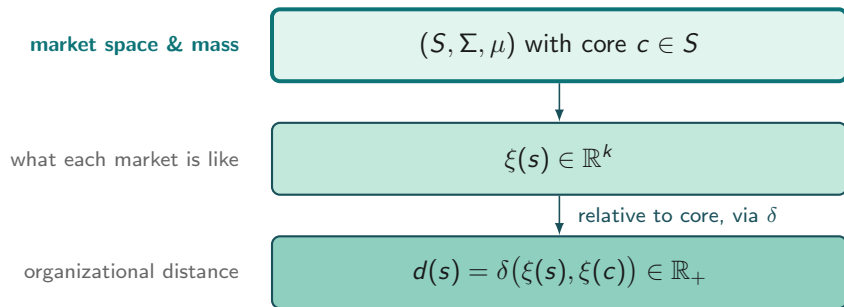
Environment

A Firm in One Picture



- One **non-rival blueprint**, created at the core, deployed across a footprint of **heterogeneous markets**.
- Each market sits at organizational distance $d(s)$ from the core.
- Deploying to distant markets must be adapted; adaptation consumes **rival headquarters attention**.

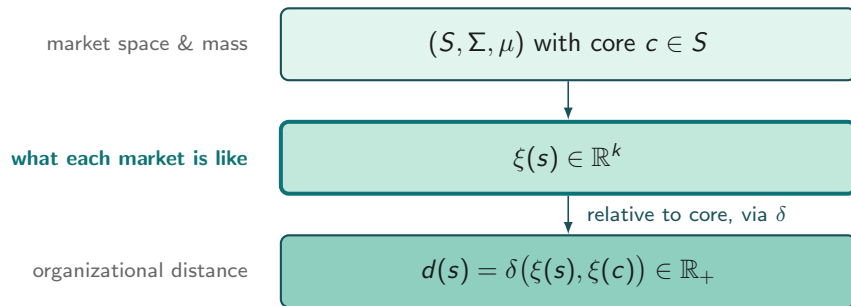
Markets and Heterogeneity



μ measures the **mass** of markets. A market s is one distinct activity the firm could run: espresso machines in Germany, cloud storage for hospitals, consumer loans to small firms.

► [details](#)

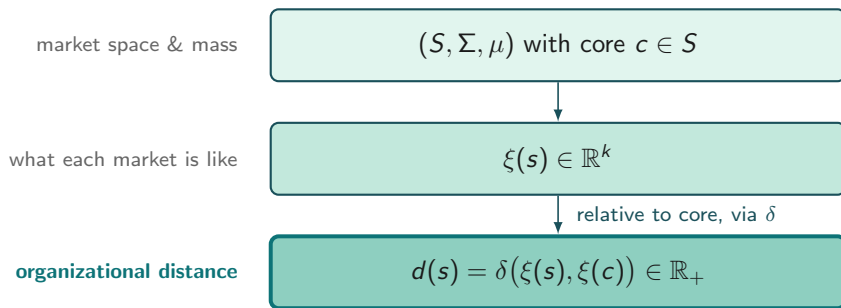
Markets and Heterogeneity



$\xi(s)$ lists the **heterogeneity-relevant attributes** of market s : geography, technology, regulation, customer composition. Each market also carries capacity $\bar{X}(s) > 0$.

► [details](#)

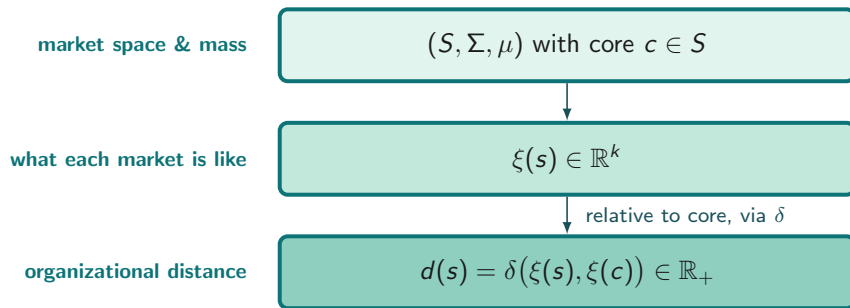
Markets and Heterogeneity



$d(s)$: a scalar summary of how far s lies from the core, for example a weighted norm $(\sum_j \omega_j |\xi_j(s) - \xi_j(c)|^p)^{1/p}$; δ need not be a norm.

► details

Markets and Heterogeneity



Why layers? Identify markets with their attributes, and more markets would mechanically mean more heterogeneity: complexity would collapse into **count**.

► details

Organizational Complexity: One Number per Firm

Integrating organizational distance over the footprint gives one number per firm:

Definition. For any measurable footprint $M \subseteq S$,

$$C(M) \equiv \int_M d(s) \mu(ds).$$

Complexity is driven by **cumulative heterogeneity**, rather than the **number** of markets served.

► [details](#)

The Blueprint and Its Deployment

- **Blueprint.** One centralized blueprint $z \geq 0$: **non-rival**, one z serves all markets at once; created at headquarters with $\chi(z)$ units of attention; adapted to each market.
- **Deployment.** Effective deployment in market s , with θ portability and A adaptation from headquarters attention $h(s)$ and local effort $\ell(s)$:

$$z_s = \underset{+}{\theta}(a(s), \underset{-}{d}(s)) z, \quad a(s) = \underset{+}{A}(\underset{+}{h}(s), \underset{+}{\ell}(s)).$$

- **Payoff.** Effective deployment and local production input $x(s)$ generate the market-level payoff $R(\underset{+}{z}_s, \underset{+}{x}(s))$.

► details

The Central Friction: Rival Headquarters Attention

- **Rival attention.** Headquarters attention is split between blueprint creation and adaptation; serving a more heterogeneous footprint makes it more expensive everywhere.
- **Cost of attention.** Every unit of attention, creation or adaptation alike, is priced at $q(C(M))$:

$$q(C(M)) \left(\underbrace{\chi(z)}_{\text{creation}} + \underbrace{\int_M h(s) \mu(ds)}_{\text{adaptation}} \right)$$

- **Congestion.** $q : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, $q(0) = 1$, $q' > 0$, $q(C) \rightarrow \infty$: an ordinary congestion wedge; the economic content is the state it congests on, cumulative heterogeneity $C(M)$, not size.

The Firm's Problem

The firm chooses blueprint quality z , footprint M , and allocations $\{x(s), \ell(s), h(s)\}$:

$$\max_{z, M, \{x, \ell, h\}} \int_M R(z_s, x(s)) \mu(ds) - q(C(M)) \left(\chi(z) + \int_M h(s) \mu(ds) \right)$$

$$\text{s.t. } x(s) + \ell(s) \leq \bar{X}(s) \quad \forall s \in M$$

$$\text{where } z_s = \theta(a(s), d(s)) z$$

$$M = \{s : x(s) + \ell(s) + h(s) > 0\}$$

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Revenues: the deployed blueprint earns $R(z_s, x(s))$ in every market of the footprint.

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The attention bill: creation plus adaptation, every unit priced at $q(C(M))$; expanding M raises the price on **all** of it.

Baseline Results

Endogenous Scope Limit

Fix capability z . What bounds the firm's reach?

Proposition (bounded complexity). At any (minimal-support) optimum,

$$C(M^*(z)) = \int_{M^*(z)} d(s) \mu(ds) < \infty.$$

- Bounded **cumulative heterogeneity** but **not** the mass $\mu(M^*)$, which may be infinite.
- A finite mass of **materially distinct** markets ($d \geq \tau$); an unbounded mass of near-core ones.

► [details](#)

Expansion Is Non-Separable

Let $\Pi_s(z; C)$ be the value of operating market s at complexity C .

Proposition (organizational congestion). $\Pi_s(z; C)$ is weakly decreasing in C ; hence adding markets Δ weakly lowers every incumbent:

$$\Pi_s(z; C(M \cup \Delta)) \leq \Pi_s(z; C(M)) \quad \forall s \in M.$$

- A market's value depends on the **whole footprint**, through C . Thus selection is non-separable.
- Separable theories (Lucas–Garicano–Melitz) **cannot generate this**.

► details

Related Expansion: A Cutoff in Heterogeneity

Which markets does the firm serve?

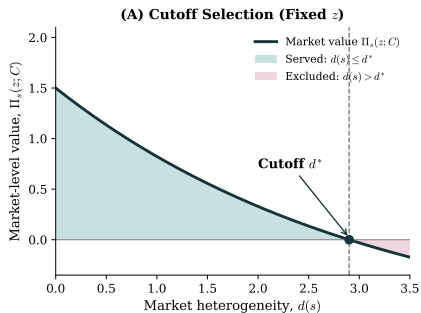
Proposition (cutoff selection). Under a heterogeneity-ordering condition, there is a threshold d^* such that the firm serves exactly the markets within it:

$$M^*(z) = \{ s : d(s) \leq d^* \}.$$

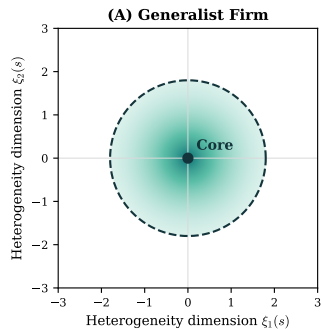
- Firms serve markets **near the core** and exclude those beyond the threshold d^* (related expansion, Hoberg & Phillips 2025).
- **No fixed costs or decreasing returns**: selection comes from **organizational scarcity**.

► details

The Cutoff and the Core–Periphery Allocation



Markets within d^* are served, those beyond are excluded.

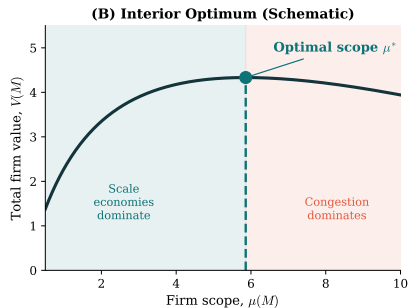
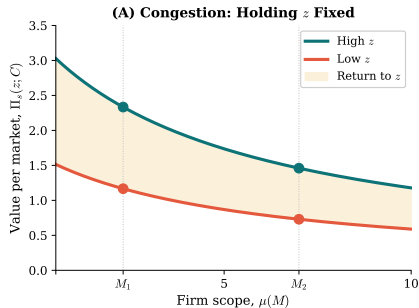


Inside the footprint, headquarters attention $h^*(s)$ declines with distance from the core (darker: more attention).

► details

Two Forces Shape Firm Scope

Now let the firm choose z . The blueprint is **non-rival**, so the return to improving it **rises with the served set**; expanding $M \rightarrow M' = M \cup \Delta$ pits this against congestion. ► details



$$\Delta \Pi = \underbrace{[\Pi_s(z^*(M'); C(M')) - \Pi_s(z^*(M); C(M'))]}_{\text{blueprint adjustment } (\geq 0)} + \underbrace{[\Pi_s(z^*(M); C(M')) - \Pi_s(z^*(M); C(M))]}_{\text{organizational congestion } (\leq 0)}$$

Complexity versus Count-Based Theories

Economy P: A Parametrized Environment

- **Markets.** A characteristic line with density f , $0 < f_{\min} \leq f \leq f_{\max}$; core c ; $d(s) = |s - c|$.
- **Footprint.** A ball of reach D , with $\rho_c(d) = f(c + d) + f(c - d)$ markets at distance d :

$$\underbrace{N(D; c) = \int_0^D \rho_c(u) du,}_{\text{count of markets served}}$$

$$\underbrace{C(D; c) = \int_0^D u \rho_c(u) du,}_{\text{complexity of the footprint}}$$

$$\underbrace{\bar{d} = C/N}_{\text{mean distance}}.$$

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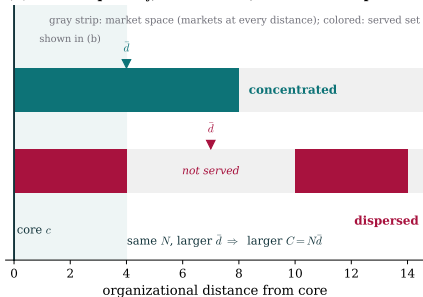
- **Payoff and capacity.** $R(z_s, x) = z_s x$, $\bar{X} \equiv 1$.
- **Adaptation and portability.** $A(h, \ell) = h$, $\theta(a, d) = (1 - e^{-a}) e^{-d/\lambda}$.
- **Congestion.** $q(C) = (1 + \kappa C)^\phi$: the price of attention rises with complexity, $C = N\bar{d}$, so it depends on **which** markets the firm serves, not only on how many.

► details

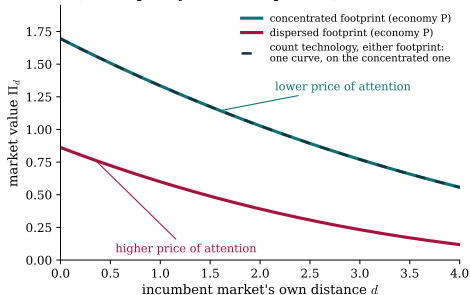
Separating Complexity from Count: Lucas, Melitz, Garicano

Hold z and the count N fixed; compare two **feasible** footprints that differ only in composition.

(A) Same Capability, Same Count, Different Composition



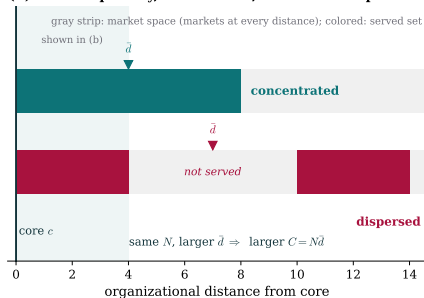
(B) Complexity Prices Composition; Count Cannot



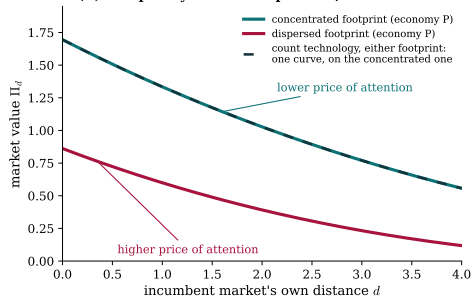
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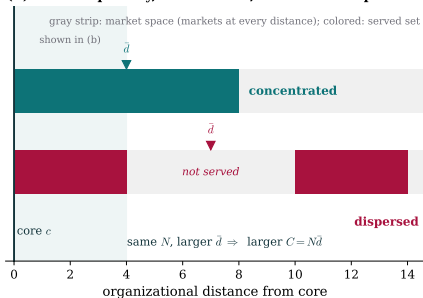
(A) The dispersed footprint places half its markets far from the core: larger mean distance $\bar{d}$, hence larger complexity $C = N\bar{d}$ at the same count.

► Formal statement

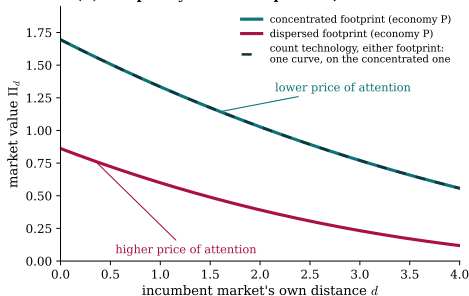
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(B) Complexity Prices Composition; Count Cannot



(B) Larger C raises the price of attention $q(C)$: every incumbent market is worth less. A cost that depends on N alone gives both firms **one price**.

► Formal statement

From Capability to Reach

Walk outward: each next market is **worth less**, and attention gets **more expensive**,

$$\underbrace{z e^{-d/\lambda}}_{\text{MB: what the market at distance } d \text{ brings}} \quad \text{against} \quad \underbrace{q(C)}_{\text{MC: the price of attention}} .$$

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3. **Fully internalized**, the exact reach $D^*(z; c)$: the next market must also cover the **rise in the bill** on everything behind; the firm stops earliest,

$$D^*(z; c) < \Phi_0(z; c) < \lambda \log z,$$

a margin absent in count-based theories.

Quantitative Illustration

Calibration: Matching Hoberg–Phillips (2025) Findings

Targeted moment	HP	Model
Mean scope	7.546	7.566
Median scope	6.000	6.124
Std. dev. of scope	5.800	5.653
Mean scope, size quintile 1	5.870	5.809
Mean scope, size quintile 2	6.890	6.595
Mean scope, size quintile 3	7.610	7.241
Mean scope, size quintile 4	8.100	8.058
Mean scope, size quintile 5	9.260	10.127
Corr(scope, log assets)	0.291	0.279
Std. dev. of log assets	2.130	2.129

D2V-Scope: the number of the 300 text-based product markets whose text matches the firm's 10-K description.

► parameters, full table ► identification ► fit

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The average firm serves 7.5 of 300 markets: scope is **tightly bounded**.

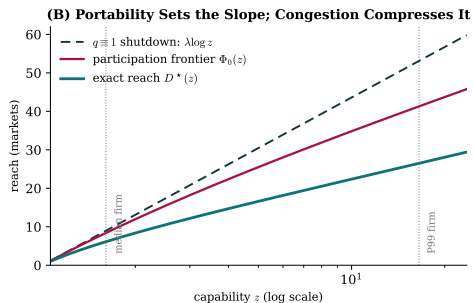
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Across size quintiles, mean scope rises only from 5.9 to 9.3: size buys **little scope**.

The Anatomy of Compression

The fitted **Pareto** capability distribution, stopped by the walk's **three cases**: no congestion ($q \equiv 1$), priced at participation (Φ_0), fully internalized (D^*).

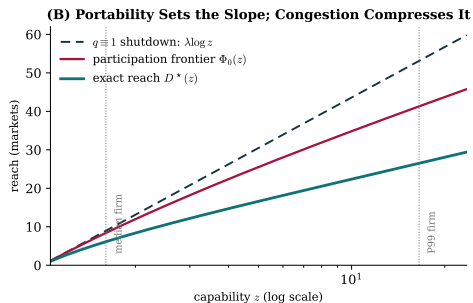


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Std. dev. of scope	11.3	9.0	5.7
Scope at 10× capability	5.9×	4.9×	4.3×
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► Tail compression ► Lucas, Melitz ► details

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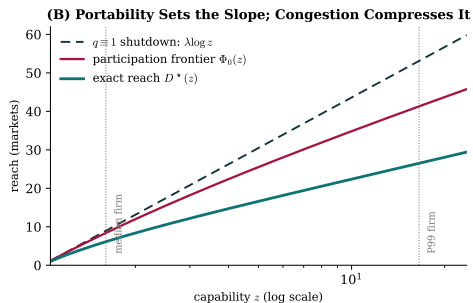
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Complexity cuts mean scope by **38%** and halves its dispersion.

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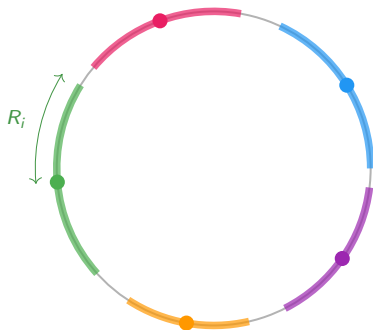
The far tail is cut **twentyfold**: the missing giants.

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Local Dominance & Concentration

A Multi-Firm Environment

Embed many firms on a shared space: the torus $\mathbb{T}_L$, a circle of circumference L .



- Core c_i (**dot**) and footprint $M_i = \{s : \delta(s, c_i) \leq R_i\}$ (**arc**).
- **Scope radius** $R_i = D^*(z_i; c_i)/2$, half the exact reach, so the arc's length is the firm's scope.
- Each firm **dominates locally** near its core.

Does this local dominance add up to **aggregate concentration**?

► [details](#)

The Geometry of Concentration

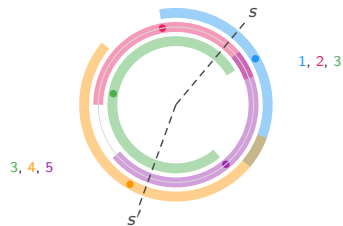
Same cores, same blueprint qualities; only the radius mapping differs.

Organizational complexity



compressed arcs, little overlap

Without scope compression



wide arcs, same firm spans s, s'

- Compressed arcs: fewer rivals at each slice, and less same-firm presence at a distance.
- Concentration within a slice and presence across slices: one object measures both.

Local and Aggregate Concentration: One Object

Firm i 's local activity share is $\omega_i(s)$. The **overlap kernel**:

$$\mathcal{O}(x) = \frac{1}{L} \int_{\mathbb{T}_L} \sum_i \omega_i(s) \omega_i(s+x) ds$$

Proposition (overlap-kernel identity).

$$\mathcal{O}(0) = \overline{\text{HHI}}_{\text{loc}} \equiv \frac{1}{L} \int_{\mathbb{T}_L} \sum_i \omega_i(s)^2 ds,$$

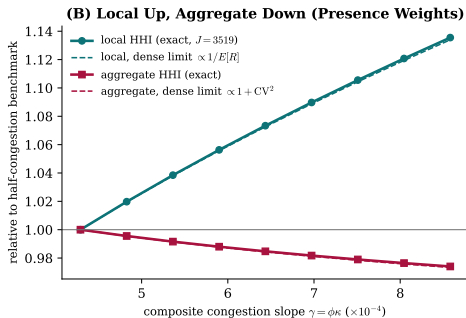
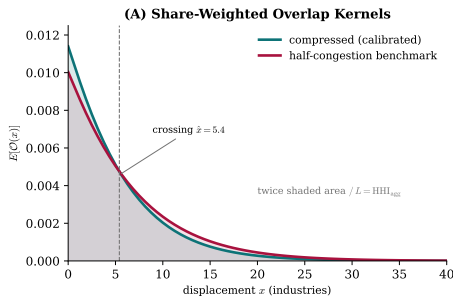
$$\frac{1}{L} \int_{\mathbb{T}_L} \mathcal{O}(x) dx = \text{HHI}_{\text{agg}} \equiv \sum_i \bar{\omega}_i^2.$$

where $\bar{\omega}_i = \frac{1}{L} \int_{\mathbb{T}_L} \omega_i$ is the average share, and HHI is the Herfindahl–Hirschman index.

- How fast $\mathcal{O}$ decays governs the gap between **local** and **aggregate** concentration.

Compression Raises Local, Lowers Aggregate Concentration

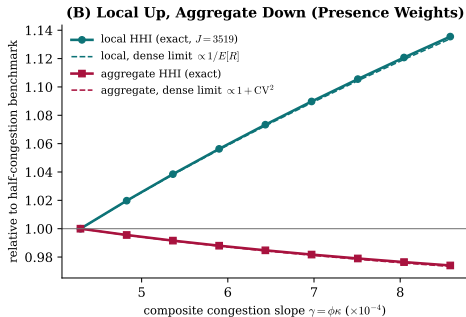
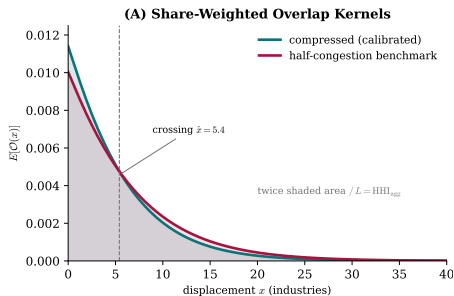
Local concentration is the **level** of scope, aggregate its **dispersion**. Complexity lowers both: local dominance rises, aggregate concentration falls.



► Deployment weights ► Formal statement

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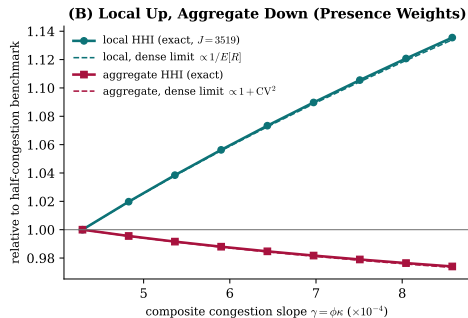
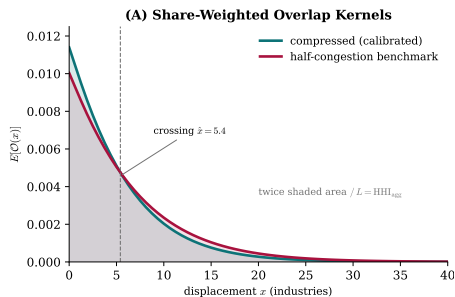


(A) Share-weighted overlap kernels, calibrated against half congestion: compression moves weight toward **short displacements**; the kernels cross at 0.7 of a mean footprint.

► **Deployment weights** ► **Formal statement**

Compression Raises Local, Lowers Aggregate Concentration

Local concentration is the **level** of scope, aggregate its **dispersion**. Complexity lowers both: local dominance rises, aggregate concentration falls.



(B) Strengthening congestion raises local HHI, +13.6% at the calibration, while the presence-weighted aggregate HHI falls, **-2.6%**.

► Deployment weights ► Formal statement

Key Takeaways

1. **An alternative to count-based theories of the firm**: scope is bounded by organizational complexity, the cumulative heterogeneity of the footprint.
2. **Non-separable, related expansion**: each added market raises the price of attention everywhere; firms grow along related lines and stop where complexity bites.
3. **Separation and compression**: composition carries its own price; Pareto capability in, exponentially bounded scope out: the **missing giants**.

Appendix: Assumptions, Closed Forms, and Tables

Assumption: Non-atomic Market Space

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Assumption (Non-atomic market space). (S, Σ, μ) is non-atomic: for every $E \in \Sigma$ with $\mu(E) > 0$ and every $\varepsilon \in (0, \mu(E))$, there exists $E' \subseteq E$, $E' \in \Sigma$, with $\mu(E') = \varepsilon$.

Rules out indivisible markets of positive mass: the firm can expand or contract its footprint by arbitrarily small amounts. The market space behaves like a continuum (Lebesgue on an interval), the natural idealization for scope spanning many small, related activities.

Assumption: Positive Heterogeneity

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Assumption (Positive heterogeneity). $d(s) > 0$ for μ -a.e. $s \in S$.

Rules out a positive-mass cluster of markets identical to the core. The normalization $d(c) = 0$ is consistent with this since the core is a null reference point. The firm may still operate across vast collections of nearly core-aligned markets with arbitrarily small $d(s)$.

Example (organizational distance). A natural form is a weighted norm,

$$d(s) = \left(\sum_{j=1}^k \omega_j |\xi_j(s) - \xi_j(c)|^p \right)^{1/p},$$

but no result requires $d(\cdot)$ to satisfy the axioms of a norm.

Assumptions: Deployment and Effectiveness

◀ back

Assumption (Adaptation and portability). $A : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ continuous, weakly increasing in each argument. $\theta : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow [0, 1]$ continuous, weakly increasing in a , weakly decreasing in d , with

$$\lim_{a \rightarrow \infty} \theta(a, d) = \bar{\theta}(d) \quad \text{for all } d \geq 0,$$

where $\bar{\theta} : \mathbb{R}_+ \rightarrow (0, 1]$ is continuous, weakly decreasing, and $\bar{\theta}(0) = 1$.

Assumption (No free deployment). For any s with $d(s) > 0$, $\theta(A(0, \ell), d(s)) = 0$ for all $\ell \geq 0$.

Assumption (Blueprint effectiveness). $R(z_s, x)$ weakly increasing in each argument.

An Inada-type portability technology: zero deployment without headquarters attention, deployment up to the ceiling $\bar{\theta}(d)$, equal to one at the core, in the limit of unbounded adaptation. A unit limit $\bar{\theta} \equiv 1$ would be incompatible, on an unbounded market space, with conditional integrability and with the single-crossing condition behind the core-periphery result; economy P uses $\bar{\theta}(d) = e^{-d/\lambda}$.

Assumptions: Congestion and Integrability

◀ back

Assumption (Organizational congestion). $q : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ with $q(0) = 1$; continuously differentiable and strictly increasing; $\lim_{C \rightarrow \infty} q(C) = \infty$.

Broader, more heterogeneous scope raises coordination burdens, increasing the marginal cost of headquarters attention anywhere in the firm, including blueprint creation $\chi(z)$.

Assumption (Conditional integrability). For fixed blueprint quality z , gross payoffs are dominated by some $\bar{R}_z \in L^1(S, \mu)$, ruling out unbounded payoffs at fixed technology.

Appendix: Support, Zero Deployment, and Materially Distinct Markets

◀ back

Served markets. The footprint is the support of activity, $M = \{s : x(s) + \ell(s) + h(s) > 0\}$ up to null sets; among payoff-equivalent optima, a **minimal support** is selected: a market is included only when excluding it would lower the objective.

Assumption (No positive payoff without effective deployment). For any market s and feasible $x \leq \bar{X}(s)$, $R(0, x) \leq 0$: favorable local conditions alone generate no rent. Used only for the zero-blueprint case of the bounded-complexity proposition.

Corollary (finite mass of materially distinct markets). For any resolution $\tau > 0$ (an ex-post slicing, not a choice of the firm), with $M_\tau^* \equiv \{s \in M^* : d(s) \geq \tau\}$,

$$\mu(M_\tau^*) \leq \frac{1}{\tau} \int_{M^*} d(s) \mu(ds) = \frac{C(M^*)}{\tau} < \infty.$$

Count-based benchmark. With constant heterogeneity $d(s) \equiv \bar{d} > 0$, $C(M) = \bar{d} \mu(M)$: bounding complexity is bounding count, the span-of-control case.

Appendix: The Market-Level Value and Strict Congestion

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Definition (market-level value function). For a market s , blueprint z , and finite complexity C ,

$$\Pi_s(z; C) \equiv \sup_{x, \ell, h \geq 0, x + \ell \leq \bar{X}(s)} \left\{ R(z_s, x) - q(C) h \right\}, \quad z_s = \theta(A(h, \ell), d(s)) z;$$

the creation cost $q(C)\chi(z)$ is common across markets and handled at the outer problem. Since q is increasing and $h \geq 0$, Π_s is weakly decreasing in C : this is all the non-separability proposition uses.

When is congestion strict? The inequality is strict whenever $C(M \cup \Delta) > C(M)$, the inner problem attains its value, and every optimizer at $C(M)$ uses strictly positive headquarters attention. If an optimizer with $h = 0$ exists, a higher price of attention need not reduce the market's value.

In economy P the value of an active market is $q\psi(w)$ with $w = ze^{-d/\lambda}/q(C)$ and $\psi(w) = w - 1 - \log w$, and $\partial_q [q\psi(ze^{-d/\lambda}/q)] = -\log w < 0$ on the active set: congestion is strict in every active market.

Appendix: The Heterogeneity-Ordering Condition

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Assumption (Heterogeneity ordering). For any fixed (z, C) , the market value $\Pi_s(z; C)$ is weakly decreasing in $d(s)$:

$$d(s_1) \leq d(s_2) \implies \Pi_{s_1}(z; C) \geq \Pi_{s_2}(z; C).$$

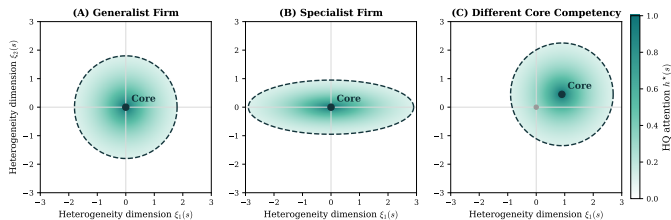
More heterogeneous markets are weakly less valuable to serve; this monotonicity is what turns selection into a threshold in d .

Sufficient primitive condition. Uniform local capacity $\bar{X}(s) = \bar{X}$: the inner feasible set is common across markets, effective deployment $\theta(A(h, \ell), d(s))$ z falls in d , so the objective $R(z_s, x) - q(C)h$ is pointwise larger at smaller d ; taking suprema delivers the ordering.

Sufficient, not necessary: the ordering also holds for non-uniform $\bar{X}$ co-varying suitably with d . It restricts neither returns to scale nor within-market technology.

Proposition: Core–Periphery Allocation

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Proposition (core–periphery allocation). Under single-crossing of the conditional gross payoff $V(h, d; z)$ in headquarters attention and heterogeneity (decreasing differences), constant local capacity, and single-valued $h^*(s)$: for any $s_1, s_2 \in M$, $d(s_1) \leq d(s_2) \implies h^*(s_1) \geq h^*(s_2)$.

Attention $h^*(s)$ over characteristic space; dashed curve: footprint boundary. Symmetric adaptation costs give a disc (a), costs accumulating slowly along ξ_1 an elongated footprint (b), a different core relocates it (c); attention declines with distance in every footprint.

Proposition: Returns to Blueprint Scale

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Proposition (returns to blueprint scale). Fix finite C . For any $z_2 \geq z_1 \geq 0$ and measurable $M_1 \subseteq M_2$,

$$\int_{M_2} [\Pi_s(z_2; C) - \Pi_s(z_1; C)] \mu(ds) \geq \int_{M_1} [\Pi_s(z_2; C) - \Pi_s(z_1; C)] \mu(ds);$$

under adaptation regularity and differentiation under the integral, also $\partial_z \int_{M_2} \Pi_s \mu(ds) \geq \partial_z \int_{M_1} \Pi_s \mu(ds)$.

Every market deploys the same blueprint, so the gain from improving it accumulates over the served set: scope and blueprint quality are complements at fixed complexity, from non-rival deployment alone. The calibration shuts this margin down ($\chi_1 = 0$) and treats capability as an exogenous primitive.

Appendix: Economy P and the Market-Level Closed Forms

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Economy P. (P1) Markets on a characteristic line with density f , core c , $d(s) = |s - c|$; distance density $\rho_c(d) = f(c + d) + f(c - d)$; ball of reach D : $N = \int_0^D \rho_c$, $C = \int_0^D u \rho_c(u) du$, $\bar{d} = C/N$ (uniform: $N = D$, $C = D^2/2$). (P2) $\bar{X}(s) \equiv 1$. (P3) $R(z_s, x) = z_s x$. (P4) $A(h, \ell) = h$, $\theta(a, d) = (1 - e^{-a}) e^{-d/\lambda}$. (P5) $q(C) = (1 + \kappa C)^\phi$. (P6) $\chi(z) = \chi_1 z^\eta$, with $\chi_1 = 0$ in the calibration. **Market-level problem.** With $w(d) \equiv z e^{-d/\lambda} / q(C)$ and $\psi(w) \equiv w - 1 - \log w$: a market at distance d is active iff $w(d) > 1$, and then

$$h^*(d) = \log w(d), \quad z_d = q(w(d) - 1), \quad \Pi_d(z; C) = q\psi(w(d)),$$

strictly decreasing in d and in C ; every market-level object depends on the rest of the footprint only through $q(C)$.

Economy P satisfies every maintained assumption of Sections 1–2, so cutoff selection and the core–periphery allocation obtain; the general density f enters only through ρ_c , and the uniform benchmark is the case with closed forms and the case calibrated.

Appendix: Count versus Complexity, Formal Statements

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Proposition 3.2 (complexity is the sufficient statistic). In economy P, fix z . For every footprint of finite complexity and every market at distance d , the market-level outcomes (price of attention, attention, deployment, value) are functions of (z, d, C) alone: the rest of the footprint matters only through $C = N\bar{d}$.

Corollary 3.1 (count versus complexity). A technology is count-based if its congestion depends on the footprint only through N . If f is constant, economy P is observationally equivalent to a count-based technology on optimal cutoff footprints, with $c(N) \equiv q(N^2/(2\rho))$. If f is not constant, no count-based technology reproduces it: firms at cores with different local density and the same count have different complexity, hence different prices of attention, under economy P, and the same price under any count-based technology.

The experiment on the slide. Uniform density, $\lambda = 10$, $\kappa = 0.02$, $\phi = 1.5$, $z = 6$; footprints $[0, 8]$ and $[0, 4] \cup [10, 14]$, $N = 8$: $C = 32$ against 56, $q(C) = 2.1$ against 3.1. Posited, not chosen: the dispersed footprint is feasible but never optimal, since the cutoff would fill its gap first.

Appendix: The Exact Optimum

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Lemma (exact optimum). The frontier is $\Phi_0(z; c) = H_0^{-1}(z; c)$ with $H_0(d; c) = e^{d/\lambda}(1 + \kappa C(d; c))^\phi$, the capability required to serve a market at distance d . For every $z > 1$ an optimal footprint exists, every minimal-support optimum is a ball, and every optimal reach $D^* \in (0, \Phi_0(z))$ satisfies

$$\Pi_{D^*}(z; C) = q'(C) D^* \left[\chi(z) + \int_0^{D^*} h^*(s) \rho_c(s) ds \right], \quad C = C(D^*),$$

equivalently $z e^{-D^*/\lambda} = q(C) w^*$ with a wedge $w^* > 1$.

The boundary market must cover the rise in the attention bill on everything the firm already does; the factor D^* is what a count-based $q(N)$ lacks, whose condition reads $\Pi_{D^*} = q'(N)[\chi + \int h^* \rho_c]$. The wedge measures how far inside the frontier the optimum sits, $0 < \Phi_0(z) - D^* \leq \lambda \log w^*$, and grows with capability: 1.14 at the median firm of the calibration, 3.08 at the ninety-ninth percentile. The optimum is unique and D^* increasing in z wherever computed (proved for $D^* \geq 2\lambda$).

Appendix: Calibrated Parameters and Targeted Moments

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<i>Targeted moments</i>	HP	Model
Mean scope	7.546	7.566
Median scope	6.000	6.124
Std. dev. of scope	5.800	5.653
Mean scope, size quintile 1	5.870	5.809
Mean scope, size quintile 2	6.890	6.595
Mean scope, size quintile 3	7.610	7.241
Mean scope, size quintile 4	8.100	8.058
Mean scope, size quintile 5	9.260	10.127
Corr(scope, log assets)	0.291	0.279
Std. dev. of log assets	2.130	2.129

<i>Parameters</i>	
Pareto tail index α_z	1.678
Portability decay λ , industries	18.94
Congestion weight κ	4.29×10^{-4}
Congestion elasticity ϕ (profile)	2.000
Size-noise dispersion σ	2.044
<i>Untargeted checks</i>	
Skewness of scope	1.58
Pr(scope > 32), data max 32	3.3×10^{-3}
Median sample max, $n = 101,535$	65
Composite slope $\gamma = \phi\kappa$	8.6×10^{-4}

Moments from Hoberg–Phillips (2025, Tables 1, 2, 5). Scope is the exact optimum $D^*(z)$ conditional on $D^* \geq 1$ ($z_1 = 1.063$); $\rho_c \equiv 1$ so reach equals scope; $\underline{z} = 1$, $\bar{X} \equiv 1$, $\chi_1 = 0$. Equal-weight squared percentage loss; every moment a closed quantile integral; optimum uniqueness and monotonicity of D^* checked at every candidate. Capability is an exogenous cross-sectional primitive; the endogenous blueprint margin is shut down, $\chi_1 = 0$.

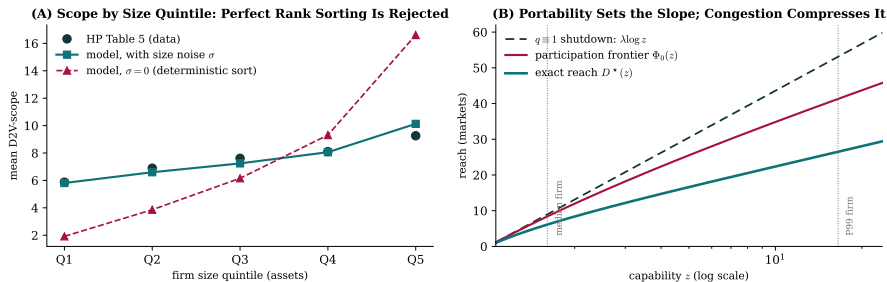
Appendix: What the Scope Distribution Identifies

◀ back

- (α_z, σ) are pinned jointly by the two size moments: $\text{sd}(\log S) = \sqrt{1/\alpha_z^2 + \sigma^2}$ exactly under the size equation, and the scope–size correlation loads on the same pair; the profiled objective over α_z is several times its minimum at $\alpha_z = 1.2$ or 2.3 .
- Without the size moments, (α_z, λ) would sit on a ridge: the frontier survival exponent $\alpha_z[m/\lambda + \phi \log(1 + \kappa m^2/2)]$ depends on four parameters through three composites. Calibrating the exact optimum helps, since the wedge depends on λ separately.
- (κ, ϕ) is the weak direction: the profiled objective falls by less than 1% as ϕ rises from 2 to 12, with κ adjusting so that $\gamma = \phi\kappa \approx 8 \times 10^{-4}$ is stable. The data pin the count–cost schedule $q(C(N)) = (1 + \kappa N^2/2)^\phi$, in effect $e^{\gamma C}$, not the pair behind it: the count–versus–complexity corollary operating inside the calibration. ϕ is fixed at the smallest value within 1% of the profile minimum.
- What would separate them is **composition**: conditional on capability, count, and own distance, incumbent outcomes vary with the mean distance of the rest of the footprint under economy P and under no count–based technology. Hoberg–Phillips tabulate operating–industry pairs by relatedness bin.

Appendix: Calibration Fit and the Compression of the Scope Mapping

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(a) Hoberg–Phillips Table 5: mean scope by asset quintile 5.87, 6.89, 7.61, 8.10, 9.26, between-quintile sd 1.14 against a total of 5.80. A deterministic size–capability relation sorts firms exactly by scope and pins a profile from 1.9 to 16.6, which the data reject; under $\log S = \log z + \sigma \varepsilon$ the quintile means are exact closed-form integrals ($\sigma = 2.04$). (b) Exact reach, participation frontier, and the portability-only mapping $\lambda \log z$ at the fitted parameters, median and 99th-percentile capabilities marked. One untargeted moment fails: the model places 3.3×10^{-3} above the observed maximum of 32, partly mechanical, D2V tops out at 32; flagged, not resolved.

Appendix: The Anatomy of Compression, Full Table

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	Portability only, $q \equiv 1$	Frontier, Φ_0	Full model, D^*
Mean scope	12.28	10.81	7.57
Median scope	8.82	8.28	6.12
Std. dev. of scope	11.28	8.96	5.65
Scope ratio at tenfold capability	5.86	4.87	4.29
Pr(scope > 32)	0.064	0.033	0.0033

All columns at the fitted (α_z, λ) and, where applicable, (κ, ϕ) , conditional on at least one market. The ratio row is $D(10z)/D(z)$ at the full model's conditional median capability $z = 1.61$; the ninety-ninth-percentile capability is 16.5, 10.3 times the median. Portability decay sets the logarithmic slope; congestion works in levels, dispersion, and the tail. Within the congestion channel, participation pricing accounts for about a quarter of the level compression in logs and internalization for three quarters: the wedge w^* is 1.14 at the median firm and 3.08 at the ninety-ninth percentile, and congestion is 4.7 and 16.7 percent of $\log H_0$ at the corresponding reaches.

Appendix: Tail Compression, Formal Statement

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Proposition 3.1 (tail compression). In economy P, for any measurable selection D^* from the optimal radii, if $\Pr(Z > x) = x^{-\alpha_z}$ for $x \geq 1$, then for every $m \geq 0$

$$\Pr(D^*(Z; c) > m) \leq \Pr(\Phi_0(Z; c) > m) = H_0(m; c)^{-\alpha_z} \leq e^{-\alpha_z m / \lambda},$$
$$\Pr(N(D^*; c) > m) \leq e^{-\alpha_z m / (\rho_{\max} \lambda)} :$$

reach and scope have exponentially bounded tails and every moment is finite, while capability is Pareto. Here $H_0(m; c) = e^{m/\lambda} (1 + \kappa C(m; c))^\phi$ and $\Phi_0 = H_0^{-1}$.

Proof. $D^* \leq \Phi_0$ by the exact-optimum lemma; $\{\Phi_0(Z; c) > m\} = \{Z > H_0(m; c)\}$ since H_0 is strictly increasing with $H_0 \geq 1$; $H_0(m; c) \geq e^{m/\lambda}$ since $C \geq 0$; and $N(D; c) \leq \rho_{\max} D$. $\square$

In the uniform benchmark the frontier tail is exact, $\Pr(\Phi_0(Z) > m) = e^{-\alpha_z m / \lambda} (1 + \kappa m^2 / 2)^{-\alpha_z \phi}$; the exact reach is asymptotically a fixed fraction of the frontier and grows at the same logarithmic rate.

Appendix: Capability and Size in the Canonical Models

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- **Span of control (Lucas 1978).** A manager of talent z hires n workers at wage w and produces $z n^\beta$, $0 < \beta < 1$:

$$\beta z n^{\beta-1} = w \implies n(z) = (\beta z / w)^{1/(1-\beta)}.$$

Pareto talent, $\Pr(Z > x) = x^{-\alpha_z}$, gives $\Pr(n(Z) > m) = (\beta/w)^{\alpha_z} m^{-\alpha_z(1-\beta)}$: size is a power of talent with elasticity $1/(1-\beta) > 1$, and the Pareto tail is passed on and thickened.

- **Monopolistic competition (Melitz 2003).** CES demand with elasticity $\sigma > 1$ gives revenue $r(\varphi) \propto \varphi^{\sigma-1}$; Pareto productivity with index k gives $\Pr(r(\varphi) > m) \propto m^{-k/(\sigma-1)}$: again a power, again a Pareto tail.
- **Economy P.** The boundary market's return decays exponentially in distance and is priced at $q(C) \geq 1$:

$$z e^{-d/\lambda} > q(C) \geq 1 \implies D^* < \Phi_0 \leq \lambda \log z, \quad \Pr(D^*(Z) > m) \leq e^{-\alpha_z m/\lambda}.$$

In each case the optimum equates the return of the last unit to its price: a return decaying polynomially ($n^{\beta-1}$, $\varphi^{\sigma-1}$) makes size a power of capability, one decaying exponentially in organizational distance ($e^{-d/\lambda}$) makes reach logarithmic, and a Pareto tail is compressed rather than passed on.

Appendix: The Torus Environment

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Characteristic space. Torus $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$ of circumference L with metric $\delta(s, s') = \min\{|s - s'|, L - |s - s'|\}$, carrying the market density f of Section 3 (periodic, bounded away from zero and infinity); lengths measured in markets, so a footprint of scope N occupies an arc of length N .

Firms. Firm i : core c_i , blueprint z_i , $d_i(s) = \delta(s, c_i)$. By the cutoff result the optimal footprint is the arc

$$M_i = \{s \in \mathbb{T}_L : \delta(s, c_i) \leq R_i\}, \quad R_i = D^*(z_i; c_i)/2,$$

half the exact reach, so that the arc's length is the firm's scope.

No double overlap: $R_i \leq L/4$, so two shifted copies of an arc overlap only on the nearest side (innocuous at the calibration: $\Pr(\text{reach} > L/2 = 150) < 10^{-10}$). **Cores** $\{c_i\}$ and $\{z_i\}$ are exogenous: no entry, exit, placement, or pricing. The overlap-kernel identity extends to any compact metric space with a transitive isometry group; the closed forms are specific to interval footprints on $\mathbb{T}_L$.

Appendix: Local Shares and the Closed Forms

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Local share (zero-activity convention): $\omega_i(s) = y_i(s)/Y(s)$ if $Y(s) > 0$, else 0; $\text{HHI}(s) = \sum_i \omega_i(s)^2$, $\overline{\text{HHI}}_{\text{loc}} = \frac{1}{L} \int \text{HHI}$, $\text{HHI}_{\text{agg}} = \sum_i \bar{\omega}_i^2$.

Benchmark. Uniform density, iid uniform cores, iid Z_i , $R_i = D^*(Z_i)/2 \leq L/4$; presence weights $y_i(s) = \mathbf{1}\{s \in M_i\}$, so $\omega_i(s) = \mathbf{1}\{s \in M_i\}/n(s)$. Write $\kappa_0 = \mathbb{E}[2R]/L$, $\kappa_x = \mathbb{E}[(2R - x)_+]/L$, $\nu = J\kappa_0$.

Lemma (closed forms). Coverage kernel $\mathcal{K}(x) = \frac{1}{L} \sum_i (2R_i - x)_+$, so $\mathbb{E}[\mathcal{K}(x)] = J\kappa_x$ and $\frac{2}{L} \int_0^{L/2} \mathbb{E}[\mathcal{K}] = \frac{4J}{L^2} \mathbb{E}[R^2]$; and

$\mathbb{E}[\mathcal{O}(x)] = \mathbb{E}[\mathcal{K}(x)] G_J(\kappa_0, \kappa_x)$, $\mathbb{E}[\overline{\text{HHI}}_{\text{loc}}] = \nu G_J(\kappa_0, \kappa_0) = \mathbb{E}\left[\frac{\mathbf{1}\{n \geq 1\}}{n}\right]$, $n \sim \text{Bin}(J, \kappa_0)$,
with $G_J(a, b) = \int_0^1 \int_0^1 [1 - 2a + b + (a - b)(u + v) + b uv]^{J-1} du dv \in (0, 1]$, increasing in b and decreasing in a .

Compression lowers κ_x (less coverage, less same-firm overlap) and lowers κ_0 , which raises G_J (fewer rivals per slice, so survivors' shares rise). Coverage dominates at long range, renormalization at short range.

Appendix: Branching and the Half-Congestion Benchmark

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Proposition (dense regime: $J \rightarrow \infty$, L and the law of R fixed). (i) $\nu \mathbb{E}[\text{HHI}(s) \mid n(s) \geq 1] \rightarrow 1$. (ii) $J \mathbb{E}[\text{HHI}_{\text{agg}}] \rightarrow 1 + \text{CV}(R)^2$. (iii) For $\mathcal{R} \leq \tilde{\mathcal{R}}$ with $\mathbb{E}[R] < \mathbb{E}[\tilde{R}]$ and $\text{CV}(R) < \text{CV}(\tilde{R})$: higher local and lower aggregate concentration under $\mathcal{R}$, and the limiting kernel ratio $Q(x) = \frac{\mathbb{E}[(2R-x)_+]}{\mathbb{E}[(2\tilde{R}-x)_+]} \left(\frac{\mathbb{E}[\tilde{R}]}{\mathbb{E}[R]} \right)^2$ exceeds one at $x = 0$ and crosses one; the crossing is a single threshold if $u \mapsto \Pr(R > u) / \Pr(\tilde{R} > u)$ is nonincreasing.

Sparse regime. $(1 - 2\nu) \mathbb{E}[\mathcal{K}(x)] \leq \mathbb{E}[\mathcal{O}(x)] \leq \mathbb{E}[\mathcal{K}(x)]$: with few firms per slice, aggregate concentration is the second moment of scope. The calibrated economy ($\nu \approx 89$) is dense; the limit formulas are within 0.2% of the finite- J values.

Half-congestion benchmark. Same capability and core draws; κ halved at fixed ϕ , so $\gamma = \phi\kappa$ falls from 8.6 to 4.3×10^{-4} . Mean scope 8.58 against 7.57, CV 0.78 against 0.75, $\nu = 100.6$ against 88.8; $\mathbb{E}[\mathcal{O}(0)]$ 0.0100 against 0.0114; HHI_{agg} 4.57 against 4.45×10^{-4} . Integrated coverage, the second moment of scope, falls by 24%: nine times the 2.6% decline of share-weighted aggregate concentration.

Appendix: Concentration under Deployment Weights

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Deployment weights $z_d = z e^{-d/\lambda} - q$ inherit the capability tail; with $\alpha_z = 1.68 < 2$ the firm-activity second moment is infinite, the $1/J$ rate fails, and the finite- J aggregate depends on the largest draws, so the object is simulated: $J = 3,519$, $L = 300$, 500 replications, common capability and core draws across economies.

Median [IQR] across replications	$q \equiv 1$	Half congestion	Calibrated
Local HHI	0.049 [0.041, 0.064]	0.058 [0.049, 0.075]	0.063 [0.054, 0.081]
Aggregate HHI ($\times 10^3$)	5.67 [4.49, 8.15]	5.79 [4.54, 8.40]	5.81 [4.58, 8.44]

Paired ratios, calibrated over half congestion: local 1.097 [1.089, 1.102], above one in every replication; aggregate 1.009 [1.004, 1.013], above one in 84% of replications, and 1.028 against $q \equiv 1$.

Presence-weighted objects on the same draws reproduce the closed forms to three digits. The concentration objects are Herfindahl indices of deployed activity, not markups or surplus; the calibration is a single pooled cross section. A coverage-only comparison, ignoring share renormalization when rivals leave a slice, overstates the aggregate decline ninefold.